Coset coordinates and the last six Higman–Sims geometry groups

Let $G = \mathrm{HS}$, with $|G| = 44\,352\,000$, and let $A = \mathrm{Aut}(G)$, with $|A| = 88\,704\,000$. We acquire the reviewed faithful models on 100 points, including $G' = A' = G$ and $C_A(G) = 1$. The fixed source is the 23-page supplement dated 6 October 2003 accompanying D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236. Its classification and the exact correspondence between source diagrams and saved parabolic families are premises of this application.

Theorem 1. *For the six specified families, the full groups are as follows. Here n is the smallest type size, a lists every model type of that size, d is the valency of the intrinsic graph, and q is the order of its full automorphism group divided by $|G|$.*

Family	*Source*	a	n	d	q	$\mathrm{Aut}(\Gamma)$	$\mathrm{Cor}(\Gamma)$
1	*2.6*	*2*	*36960*	*24*	*2*	A	A
2	*2.5*	*1*	*11550*	*64*	*2*	A	A
10	*3.27*	*2,3*	*15400*	*180*	*1*	G	A
16	*3.21*	*2,3*	*15400*	*24*	*2*	A	$A \times C_2$
17	*3.20*	*1,3*	*15400*	*24*	*2*	A	$A \times C_2$
18	*3.19*	*1*	*15400*	*24*	*2*	A	A

In the retained source, 3.20 prints $\mathrm{Aut}(\Gamma) = G$ *and* $\mathrm{Cor}(\Gamma) = A$*, while 3.21 prints* $\mathrm{Aut}(\Gamma) = G$ *and* $\mathrm{Cor}(\Gamma) = G \times C_2$*. The theorem replaces both entries in each of these two diagrams. Pages 7 and 8 were inspected directly. The four other rows agree with the printed groups, using the source convention when a correlation group is omitted. Previously recorded corrections to a type size in 2.6 and an incidence number in 2.5 are separate.*

Proving the point action from literal cosets. Fix a family $P_1, \dots, P_r$ and a smallest type a. Put $P = P_a$ and $n = [G : P]$. Let $g_1, \dots, g_t$ be the ordered generators of the original 100-point model. The certificate supplies elements $c_1, \dots, c_n \in G$ and permutations $r_1, \dots, r_t$ of $\{1, \dots, n\}$. The reader verifies

$$c_i g_j c_{i^{r_j}}^{-1} \in P \qquad (1 \le i \le n,\ 1 \le j \le t). \tag{1}$$

It also verifies transitivity of $\langle r_1, \dots, r_t \rangle$, the existence of a label b with $c_b \in P$, and $\mathrm{Core}_G(P) = 1$ in the original model.

The map $i \mapsto Pc_i$ respects each generator by (1). Every coset is reached from P by a positive generator word, since G is finite. Thus this map is surjective; its domain and codomain both have n elements, so it is bijective. Under this bijection the r_j are exactly right multiplication by g_j. Consequently they define a homomorphism $\phi : G \to E = \langle r_1, \dots, r_t \rangle$, whose kernel is $\mathrm{Core}_G(P) = 1$. The image is E by definition. Only after these checks does the program install the established homomorphism and its injectivity, surjectivity and order attributes in GAP. There is no unchecked homomorphism premise. Across the nine anchors, all 2 813 580 generator edges are checked.

Recovering objects from points. Write $E_b = \phi(P_a)$. A type-j base object has point shadow $S_j = b^{\phi(P_j)}$. Direct shadows suffice except at the two anchors of family 10, where type 1 requires additional incident objects. An encoding consists of its shadow together with typed sets of previously encoded incident objects. The reconstruction order is acyclic. For a reference to type k, the reader constructs the entire orbit of its base encoding under $\phi(P_j)$ and checks that its size is $[P_j : P_j \cap P_k]$. Induction then identifies that orbit with all incident type-k objects, with none omitted or duplicated.

Lemma 2. *Let a group E act on a set of objects equipped with an equivariant carrier in a point set. Suppose b belongs to the carrier S of an object e, and $K \leq E$ fixes e and is transitive on S. Then*

$$E_e = \langle K, (E_b)_e \rangle.$$

Proof. If $h \in E_e$, then $b^h \in S$. Choose $k \in K$ with $b^k = b^h$. The element hk^{-1} fixes both b and e, so $h \in (E_b)_e K$. The reverse inclusion follows because both factors fix e. □

For each recorded encoding the program checks its exact shadow, every referenced orbit, invariance under all generators of $\phi(P_j)$, and membership in $\phi(P_j)$ of every generator of its stabilizer inside E_b. Lemma 2 therefore gives full stabilizer $\phi(P_j)$. The orbit of the encoding has size $[G : P_j]$, so the map from type-j objects to encodings is injective. At family 10, anchor 2 reconstructs type 1 using types 2 and 3, and anchor 3 reconstructs type 1 using type 2. All remaining encodings use their direct shadows. The verifier requires every type to be reached. A typed automorphism fixing all anchor points then fixes every encoding, by induction, and hence every object. Its restriction to the anchor is faithful.

The intrinsic relation. We acquire the previously proved common-incidence formula. If O is a non-diagonal orbit of E_b on points, the number of type-j objects incident with a specified pair in the corresponding orbital is

$$m_j(O) = \frac{[G : P_j]|S_j|\,|S_j \cap O|}{n|O|}.$$

This formula counts objects, including distinct objects with equal shadows. The proof counts triples of an object and an ordered pair in its shadow; transitivity of P_j on that shadow makes the number of second points constant. For each pair take the sorted multiset of $([G : P_j], m_j(O))$ over all types. Correlations preserve it, including multiplicities. Order the distinct off-diagonal signatures lexicographically and choose the first whose relation has the indicated valency d. Every chosen relation is checked to be symmetric and irreflexive. The selection rule is intrinsic, so a correlation taking one anchor to another takes its graph to the other anchor graph. All smallest types are included.

The base neighbour set is propagated along the verified point generators. The program visits all n points and checks every adjacency row under every generator afterwards. Thus possible conflicting propagation paths cannot silently produce another graph. All rows have valency d.

Complete bounds and an explicit outer action. Let D be the full automorphism group of one of these graphs. A complete unseeded Traces calculation supplies $|D| = q|G|$ as in the table. The graph reader checks the whole adjacency array and all known E generators before calling Traces; it checks every returned generator afterwards. Only exact integer orders below 2^{53}, with zero exponent and clean completion status, are accepted. This is a computational upper bound, not a deduction from the returned generators alone.

When $q = 1$, $E \le D$ and equal orders imply $D = E$. When $q = 2$, we identify the extension directly. In the original model put $B = \bigcap_j P_j$ and $S = \bigcap_j N_A(P_j)$. The calculation verifies $S \cap G = B$, $|S| = 2|B|$, and an element $x \in S \setminus G$ with $x^2 \in P_a$. The latter element exists in each of the seven instances where $q = 2$. Starting with $b^u = b$, it propagates a point permutation u using the equations

$$(i^{\phi(g_j)})^u = (i^u)^{\phi(g_j^x)}.$$

Every encountered equation is checked, every point is reached, and the result is required to be bijective. The verifier then checks, as literal permutation identities,

$$\phi(g_j)^u = \phi(g_j^x) \quad \text{for every } j, \qquad u^2 = \phi(x^2).$$

It also checks u on every adjacency row of the actual graph.

These identities show that u normalizes E and $u^2 \in E$. Moreover, $u \notin E$. Otherwise $u = \phi(g)$ would give the same automorphism of G as x, so $xg^{-1} \in C_A(G) = 1$, contrary to $x \notin G$. Consequently $F = \langle E, u \rangle$ has order $2|G|$. Since $F \le D$ and the complete upper bound gives that same order, $D = F$. In both cases $E \lhd D$ and D/E is abelian. Perfection of G yields $E = E' \le D' \le E$, hence $D' = E$.

From graph groups to geometry groups. The preceding equality identifies the specified G intrinsically at every possible image anchor. If c is a correlation, conjugation by its point restriction carries one graph's derived group to the other. For every $g \in G$, the restriction of cgc^{-1} to the image anchor therefore equals that of some $h \in G$. The typed automorphism $h^{-1}cgc^{-1}$ fixes that anchor pointwise and is trivial by the faithful restriction just proved. Thus G is normal in the full correlation group.

The acquired coset normalizer theorem is now applicable. Let T be the setwise stabilizer of the parabolic family in A, allowing all type permutations. It gives

$$|\operatorname{Aut}(\Gamma)| = [G : B]|S|, \qquad |\operatorname{Cor}(\Gamma)| = [G : B]|T|.$$

For each type permutation, successive exact transporter intersections retain precisely the automorphisms carrying all already assigned parabolics to their targets. Positive representatives are checked on the entire family. Necessary subgroup-order tests discard only impossible permutations; no successful representative is inferred from an order.

Family	$\lvert B\rvert$	$\lvert S\rvert$	$\lvert T\rvert$	$\lvert T \cap G\rvert/\lvert B\rvert$	$\lvert \mathrm{Cor}(\Gamma)\rvert$
1	200	400	400	1	88704000
2	240	480	480	1	88704000
10	16	16	32	1	88704000
16	32	64	128	2	177408000
17	32	64	128	2	177408000
18	40	80	80	1	88704000

In each row $S \cap G = B$. The typed group therefore acts faithfully by conjugation on G and is identified with G or A by its order and its actual embedding in $\mathrm{Aut}(G)$. The full centralizer is the acquired quotient $(T \cap G)/B$. In families 16 and 17 it has order two, is central, and meets the typed group trivially. Their product has the full correlation group order, proving $A \times C_2$. In the other four rows that centralizer is trivial; conjugation faithfully embeds the full group in A and its order identifies it with A. These are identifications by explicit normalizer and centralizer arguments, not order-based group recognition.

Controls and failed computations. Literal subgroup enumeration in S_3, D_8, A_4 checks all coset edges, the carrier stabilizer lemma, and the incidence formula against direct counts, including repeated shadows. A separate example shows that carrier transitivity cannot be omitted from Lemma 2. Actual damaged certificates change a coset representative, a point generator, a shadow, and an acyclic type reference; each must be rejected. Graph controls change domains, edges and permutations. Ordinary and optimized Python must agree on all mathematical controls.

The Traces interface is additionally checked on every graph on four labelled vertices against all 24 permutations, including equality with the group generated by its output. Nonempty input generator lists are forbidden. An earlier interface that supplied known generators failed this control: for the complete graph on four vertices with one supplied transposition it reported order 18 instead of 24. That program, its input and the diagnostic are retained and supply no upper-bound premise here. All nine accepted bounds are fresh unseeded computations. Earlier capped nauty attempts, an overrestrictive preliminary outer-normalizer test and an invalid time-cap request are also retained without mathematical credit.

Attribution and remaining scope. The original geometry classification and source assertions are attributed to the retained manuscript and supplement. The coordinate correspondence, shadow reconstruction framework, common-incidence count and coset normalizer theorem are acquired from separately reviewed work. This component contributes the literal coset-edge proof, the use of a point stabilizer to test an object's full stabilizer, the explicit outer action, and their nine concrete applications. These are elementary arguments; historical novelty is not claimed. GAP's exact group algorithms and Traces's complete upper bounds remain computational premises. No Lean verification of this component is claimed.

The six full-group determinations complete this particular part of the 86-family analysis, conditional on the acquired classification. Abstract parabolic structures, the other uncompleted source clauses, and final whole-manuscript review remain separate. There is no new conjecture resolution or compression measurement in this component.